Cambridge International AS & A Level

Cambridge Assessment International Education
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MATHEMATICS **9709/13**

Paper 1 **October/November 2018**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

- The symbol FT implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.

  - Note: B2 or A2 means that the candidate can earn 2 or 0.
    B2/1/0 means that the candidate can earn anything from 0 to 2.

  The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.

- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking *g* equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF/OE Any Equivalent Form (of answer is equally acceptable) / Or Equivalent

AG Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)

CAO Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)

CWO Correct Working Only – often written by a ‘fortuitous’ answer

ISW Ignore Subsequent Working

SOI Seen or implied

SR Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1 A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting.

PA –1 This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

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Question	Answer	Marks	Guidance
1	7C5 $x^2(-2/x)^5$ soi	**B1**	Can appear in an expansion. Allow 7C2
	21×-32 soi	**B1**	Identified. Allow $(21x^2)\times(-32\,x^{-5})$. Implied by correct answer
	–672	**B1**	Allow $\frac{-672}{x^3}$. If 0/3 scored, 672 scores SCB1
		3	

Question	Answer	Marks	Guidance
2	$f'(x) = 3x^2 + 4x - 4$	**B1**	
	Factors or crit. values or sub any 2 values $(x \neq -2)$ into $f'(x)$ soi	**M1**	Expect $(x+2)(3x-2)$ or –2, ⅔ or any 2 subs (excluding $x = -2$).
	For $-2 < x < \frac{2}{3}$, $f'(x) < 0$; for $x > \frac{2}{3}$, $f'(x) > 0$ soi Allow $\leqslant, \geqslant$	**M1**	Or at least 1 specific value $(\neq -2)$ in each interval giving opp signs Or $f'(\frac{2}{3}) = 0$ and $f''(\frac{2}{3}) \neq 0$ (i.e. gradient changes sign at $x = \frac{2}{3}$)
	Neither www	**A1**	Must have 'Neither'
	ALT 1 At least 3 values of f(x)	**M1**	e.g. f(0) = 7, f(1) = 6, f(2) = 15
	At least 3 <u>correct</u> values of f(x)	**A1**	
	At least 3 <u>correct</u> values of f(x) spanning $x = \frac{2}{3}$	**A1**	
	Shows a decreasing and then increasing pattern. Neither www	**A1**	Or similar wording. Must have 'Neither'
	ALT 2 $f'(x) = 3x^2 + 4x - 4 = 3\left(x + \frac{2}{3}\right)^2 - \frac{16}{3}$	**B1B1**	Do not condone sign errors
	$f'(x) \geqslant -\frac{16}{3}$	**M1**	
	$f'(x) < 0$ for some values and > 0 for other values. Neither www	**A1**	Or similar wording. Must have 'Neither'
		4	

Question	Answer	Marks	Guidance
3(i)	0.8 oe	**B1**	
		1	
3(ii)	$BD = 5\sin \textit{their}\ 0.8$	**M1**	Expect 3.58(7). Methods using degrees are acceptable
	$DC = 5 - 5\cos \textit{their}\ 0.8$	**M1**	Expect 1.51(6)
	Sector = ½ × 5^2 × *their*0.8 OR Seg = ½ × 5^2 × [*their* 0.8 – sin*their* 0.8]	**M1**	Expect 10 for sector. Expect 1.03(3) for segment
	Trap = ½(5 + *theirDC*) × *theirBD* oe OR Δ*BDC* = ½*theirBD* × *theirCD*	**M1**	OR (for last 2 marks) if *X* is on *AB* and *XC* is parallel to *BD*:
	Shaded area = 11.69 – 10 OR 2.71(9) – 1.03(3) = 1.69 cao	**A1**	*BDCX* –(sector – Δ*AXC*) = 5.43(8) – [10 – 6.24(9)] = 1.69 cao M1A1
		5	

Question	Answer	Marks	Guidance
4(i)	Gradient, *m*, of *AB* = 3/4	**B1**	
	Equation of *BC* is $y - 4 = \frac{-4}{3}(x-3)$	**M1A1**	Line through (3, 4) with gradient $\frac{-1}{m}$ (M1). (Expect $y = \frac{-4}{3}x + 8$)
	$x = 6$	**A1**	Ignore any *y* coordinate given.
		4	

Question	Answer	Marks	Guidance
4(ii)	$(AC)^2 = 7^2 + 1^2 \rightarrow AC = 7.071$	M1A1	M mark for $\sqrt{(their 6 + / -1)^2 + 1}$.
		2	

Question	Answer	Marks	Guidance
5	$a + (n-1)3 = 94$	B1	
	$\frac{n}{2}[2a + (n-1)3] = 1420$ OR $\frac{n}{2}[a + 94] = 1420$	B1	
	Attempt elimination of a or n	M1	
	$3n^2 - 191n + 2840\ (=0)$ OR $a^2 - 3a - 598\ (=0)$	A1	3-term quadratic (not necessarily all on the same side)
	$n = 40$ (only)	A1	
	$a = -23$ (only)	A1	Award 5/6 if a 2nd pair of solutions (71/3, 26) is given in addition or if given as the only answer.
		6	

Question	Answer	Marks	Guidance
6	$(\mathbf{BO}) = -8\mathbf{i} - 6\mathbf{j}$	**B1**	OR (**OB**) = 8**i** + 6**j**
	$(\mathbf{BF}) = -6\mathbf{j} - 8\mathbf{i} + 7\mathbf{k} + 4\mathbf{i} + 2\mathbf{j} = -4\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$	**B1**	OR (**FB**) = 4**i** + 4**j** – 7**k**
	$(\mathbf{BF}.\mathbf{BO}) = (-4)(-8) + (-4)(-6)$	**M1**	OR (**FB.OB**) Expect 56. Accept one reversed but award final A0
	$\lvert\mathbf{BF}\rvert \times \lvert\mathbf{BO}\rvert = \sqrt{4^2 + 4^2 + 7^2} \times \sqrt{8^2 + 6^2}$	**M1**	Expect 90. At least one magnitude <u>methodically</u> correct
	Angle $OBF = \cos^{-1}\left(\frac{their\,56}{their\,90}\right) = \cos^{-1}\left(\frac{56}{90}\right)$ or $\cos^{-1}\left(\frac{28}{45}\right)$	**DM1A1**	Or equivalent 'integer' fractions. All M marks dependent on use of (±)**BO** and (±)**BF**. 3rd M mark dep on both preceding M marks
		6	

Question	Answer	Marks	Guidance
7(i)	$\frac{(\tan\theta + 1)(1 - \cos\theta) + (\tan\theta - 1)(1 + \cos\theta)}{(1 + \cos\theta)(1 - \cos\theta)}$ soi	**M1**	
	$\frac{\tan\theta - \tan\theta\cos\theta + 1 - \cos\theta + \tan\theta - 1 + \tan\theta\cos\theta - \cos\theta}{1 - \cos^2\theta}$ www	**A1**	
	$\frac{2(\tan\theta - \cos\theta)}{\sin^2\theta}$ www **AG**	**A1**	
		3	

Question	Answer	Marks	Guidance
7(ii)	$(2)(\tan\theta - \cos\theta)\ (=0) \rightarrow (2)\left(\frac{\sin\theta}{\cos\theta} - \cos\theta\right)\ (=0)$ soi	**M1**	Equate numerator to zero and replace $\tan\theta$ by $\sin\theta / \cos\theta$
	$(2)\left(\sin\theta - \left(1 - \sin^2\theta\right)\right)\ (=0)$	**DM1**	Multiply by $\cos\theta$ and replace $\cos^2\theta$ by $1 - \sin^2\theta$
	$\sin\theta = 0.618(0)$ soi	**A1**	Allow (√5–1)/2
	$\theta = 38.2°$	**A1**	Apply penalty –1 for extra solutions in range
		4	

Question	Answer	Marks	Guidance
8(i)	$y = \frac{1}{3}ax^3 + \frac{1}{2}bx^2 - 4x\ \ (+c)$	**B1**	
	$11 = 0 + 0 + 0 + c$	**M1**	Sub $x = 0$, $y = 11$ into an integrated expression. c must be present
	$y = \frac{1}{3}ax^3 + \frac{1}{2}bx^2 - 4x + 11$	**A1**	
		3	
8(ii)	$4a + 2b - 4 = 0$	**M1**	Sub $x = 2$, $dy/dx = 0$
	$\frac{1}{3}(8a) + 2b - 8 + 11 = 3$	**M1**	Sub $x = 2$, $y = 3$ into an integrated expression. Allow if 11 missing
	Solve simultaneous equations	**DM1**	Dep. on both M marks
	$a = 3$, $b = -4$	**A1A1**	Allow if no working seen for simultaneous equations
		5	

Question	Answer	Marks	Guidance
9(i)	For *their* 3-term quad a recognisable application of $b^2 - 4ac$	**M1**	Expect $2x^2 - x(3+k) + 1 - k^2 \quad (=0)$ oe for the 3-term quad.
	$\left(b^2 - 4ac =\right)\ (3+k)^2 - 4(2)(1-k^2)$ oe	**A1**	Must be correct. Ignore any RHS
	$9k^2 + 6k + 1$	**A1**	Ignore any RHS
	$(3k+1)^2 \geqslant 0$ Do not allow > 0. Hence curve and line meet. **AG**	**A1**	Allow $(9)\left(k + \frac{1}{3}\right)^2 \geqslant 0$. Conclusion required.
	ALT Attempt solution of 3-term quadratic	**M1**	
	Solutions $x = k+1, \quad ½(1-k)$	**A1A1**	
	Which exist for all values of k. Hence curve and line meet. **AG**	**A1**	
		4	

Question	Answer	Marks	Guidance
9(ii)	$k = -1/3$	**B1**	**ALT** $\mathrm{d}y/\mathrm{d}x = 4x-3 \Rightarrow 4x-3=k$
	Sub (one of) *their* $k = -\frac{1}{3}$ into either line 1 $\rightarrow$ $2x^2 - \frac{8}{3}x + \frac{8}{9}\ (=0)$ Or into the derivative of line 1 $\rightarrow 4x\ -(3+k)(=0)$	**M1**	Sub $k = 4x-3$ into line 1 $\rightarrow$ $2x^2 - x(4x) + 1 - (4x-3)^2\ (=0)$
	$x = 2/3$ Do not allow unsubstantiated $\left(\frac{2}{3}, -\frac{1}{9}\right)$ following $k = -\frac{1}{3}$	**A1**	$x = 2/3$, $y = -1/9$ (both required) [from $-18x^2 + 24x - 8$ (=0) oe]
	$y = -1/9$ Do not allow unsubstantiated $\left(\frac{2}{3}, -\frac{1}{9}\right)$ following $k = -\frac{1}{3}$	**A1**	$k = -1/3$
		4	

Question	Answer	Marks	Guidance
10(i)	$V = 4(\pi)\int(3x-1)^{-2/3}\,\mathrm{d}x \quad = 4(\pi)\left[\frac{(3x-1)^{1/3}}{1/3}\right][\div 3]$	**M1A1A1**	Recognisable integration of y^2 (M1) Independent A1, A1 for [ ] [ ]
	$4(\pi)[2-1]$	**DM1**	Expect $4(\pi)(3x-1)^{1/3}$
	4π or 12.6	**A1**	Apply limits $\frac{2}{3} \rightarrow 3$. Some working must be shown.
		5	

Question	Answer	Marks	Guidance
10(ii)	$\mathrm{d}y/\mathrm{d}x = (-2/3)(3x-1)^{-4/3} \times 3$	**B1**	Expect $-2(3x-1)^{-4/3}$
	When $x = 2/3,\ y = 2$ soi $\mathrm{d}y/\mathrm{d}x = -2$	**B1B1**	2nd B1 dep. on correct expression for d*y*//d*x*
	Equation of normal is $y - 2 = ½(x - ⅔)$	**M1**	Line through (⅔, *their* 2) and with grad $-1/m$. Dep on m from diffn
	$y = \frac{1}{2}x + \frac{5}{3}$	**A1**	
		5	

Question	Answer	Marks	Guidance
11(i)	$[2]\left[(x-3)^2\right][-7]$	**B1B1B1**	
		3	
11(ii)	Largest value of k is 3. Allow (k =) 3.	**B1**	Allow $k \leqslant 3$ but not $x \leqslant 3$ as final answer.
		1	

Question	Answer	Marks	Guidance
11(iii)	$y=2(x-3)^2-7 \to (x-3)^2=\frac{1}{2}(y+7)$ or with x/y transposed	**M1**	Ft *their a*, *b*, *c*. Order of operations correct. Allow sign errors
	$x=3\pm\sqrt{\frac{1}{2}(y+7)}$ Allow 3 + √ or 3 –√ or with x/y transposed	**DM1**	Ft *their a*, *b*, *c*. Order of operations correct. Allow sign errors
	$f^{-1}(x)=3-\sqrt{\frac{1}{2}(x+7)}$	**A1**	
	(Domain is x) $\geqslant$ *their* – 7	**B1FT**	Allow other forms for interval but if variable appears must be x
		4	
11(iv)	$x+3\leqslant 1$. Allow $x + 3 = 1$	**M1**	Allow $x+3\leqslant k$
	largest p is -2 . Allow (p =) –2	**A1**	Allow $p\leqslant -2$ but not $x\leqslant -2$ as final answer.
	$fg(x)=f(x+3)=2x^2-7$ cao	**B1**	
		3	